Problem 7.2

What deposit made today will provide for a payment of $1,000 in 1 year and $2,000 in 3 years, if the effective rate of interest is 7.5%?

Solution

Let $v = \dfrac{1}{1.075}$ be the discount factor.

The deposit required today is the present value of the two future payments:

$$X = 1000v + 2000v^3.$$

Substituting $v = \dfrac{1}{1.075}$,

$$X = 1000\left(\frac{1}{1.075}\right) + 2000\left(\frac{1}{1.075}\right)^3.$$

Now compute each term:

$$1000\left(\frac{1}{1.075}\right) \approx 930.2326$$

and

$$2000\left(\frac{1}{1.075}\right)^3 \approx 1609.9211.$$

Therefore,

$$X \approx 930.2326 + 1609.9211 = 2540.1537.$$

Hence,

$$\boxed{\$2{,}540.1537}.$$